Neural Processing Letters (2023) 55:3827–3842
https://doi.org/10.1007/s11063-020-10272-1

# Securing Blockchain Transactions Using Quantum Teleportation and Quantum Digital Signature

Sheetal Singh[1] · Nikhil Kumar Rajput[1] · Vipin Kumar Rathi[1] · Hari Mohan Pandey[2] · Amit Kumar Jaiswal[3] · Prayag Tiwari[4]

Published online: 8 June 2020


**Abstract**

Blockchain is the new disruptive technology which is gaining momentum in several domains of real-world applications such as Bitcoin—the most well-known example, and primarily in the financial sector. Distributed ledger system which is the main feature of blockchain technology protected across harmful updates using cryptographic techniques, e.g. hashing and digital signatures, is spread over the network so that no one is the owner of the ledger. This poses a significant challenge in the areas of performance, security, and privacy using this emerging technology. We propose an algorithm to provide a secure mechanism for performing transactions on the Blockchain network by adopting quantum digital signatures (QDS) and quantum teleportation phenomenon of quantum computing. The proposed algorithm uses key pairs to generate private keys and corresponding public keys; the quantum digital signatures are used to sign the message which are then distributed over the blockchain network using the teleportation phenomenon. The security of the keys is dependent on the fundamental principles of quantum mechanics that doesn't allow forging. The Einstein–Podolsky–Rosen (EPR) pair of particles is used to communicate the quantum information via a quantum channel for teleportation, which is generated by operating Bell measurements with corresponding EPR pairs. The original state of the particle (sender) is destroyed once the information is transported to the other particle (receiver) and the QDS scheme also validates the qubits received. The twofold validation process thereby provides high-level security in the transactions.

**Keywords** Blockchain · Quantum entanglement · Quantum digital signatures · Quantum teleportation · Cryptocurrencies

## 1 Introduction

The notion of establishing a peer to peer transaction system thereby removing the third party, generally being the financial institutions stems from the proposed electronic transaction system in [1], globally known as Bitcoin. The currency allows the transactions without any central authority because of its distributed nature. There have been several other attempts also for building digital currencies, but most of them required a central authority for validation [2]. Blockchain is the central idea behind the conceptualisation of Bitcoin. It is a decentralized

Extended author information available on the last page of the article ledger that maintains the record of transactions using cryptographic methods. However, there are certain elementary risks associated with distributed digital currencies such as double spend, Sybil and Eclipse attack and DoS (Denial of Service) to name a few [3]. Recent work [4] on security issues in internet of things (IOT) [5–8] has been explored using blockchain.

A novel mechanism to enhance security in blockchain technology using the competence of quantum computing has been proposed in this work. The advantage lies in the coherent superposition of a large number of states that enables massive parallel computations in a quantum environment. The most substantial and mysterious concept of entanglement has made way for several new modes of communication including quantum teleportation, super-dense coding, quantum key distribution, and quantum cryptography. The existing security of blockchain technology is based on public key infrastructure that uses asymmetric public and private keys. We intend to use quantum digital signature scheme rather than the classical digital signatures where a sender can generate only a limited number of public keys from the given private keys. The Quantum Digital Signature scheme is discussed in detail in further section of this paper. The QDS requires a quantum channel for communication of qubits from source to destination. This communication is attained by quantum teleportation in which an unknown quantum state is transmitted from one particle to another through a quantum channel. The original quantum channel is represented by a Bell state that is an EPR pair.

### 1.1 Contributions

The main contribution of this study can be encapsulated as follows:

1. We propose to apply quantum digital signatures on the existing blockchain technology to intensify the privacy and security of the user's information.
2. We propose to leverage the quantum teleportation mechanism within the blockchain network to minimize the latency and security issues.
3. We utilize the Einstein–Podolsky–Rosen paradox to develop an algorithm for securing transactions which is a major component in Blockchain network.

### 1.2 Organization

The remaining part of this paper is connected as follows. In Sect. 2, we present a brief overview of Blockchain and a number of case studies demonstrating the thefts and hacks executed in the recent times on the cryptocurrencies using this technology. Sections 3 and 4 gives an insight of quantum computing and quantum entanglement under-laying a detailed exploration of quantum teleportation. The next section describes the quantum digital signature scheme followed by the methodology used for the proposed algorithm in Sect. 6. Section 7 presents the Quantum Digital Signature (QDS) based quantum teleportation algorithm followed by the discussion section which elaborates the significance of our research from the viewpoint of the existing solutions. In Sect. 9, we present the future work and conclude our work.

## 2 Blockchain Technology

Blockchain is a distributed public ledger that records all the transactions for every single unit of digital currency. Since there is no central authority for the verification of transactions; hence

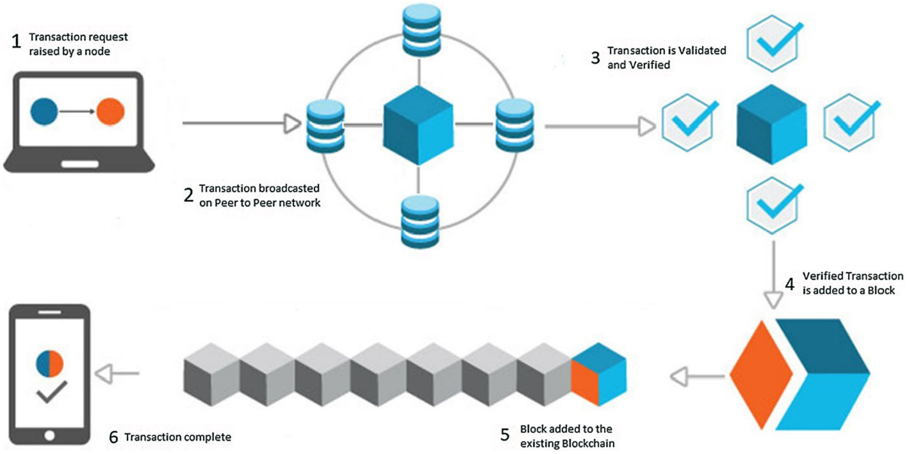



**Fig. 1** Transaction flow in a Blockchain

it is decentralized (Fig. 1). The security in the blockchain is established by cryptographic proof where various nodes in the network verify the blocks of transactions for their validation in a repeated frame of time. The member nodes who agree to contribute their resources to the system can earn rewards for verifying transactions by successfully solving the cryptographic puzzles. The decentralized nature of the blockchain technology requires the miner nodes to be genuine so that the security of network is integrated [9].

### 2.1 Security in Blockchain

Present Blockchain technology based systems extensively uses public key cryptography which is primarily based on ECDSA (Elliptic Curve Digital Signature Algorithm) that essentially uses a pair of keys comprising of a public key and a private key. While public keys are publicly known and essential for identification, the private keys are stored secretly and are used for encryption and authentication [3,9]. These keys are kept in a cryptocurrency wallet from where one can perform a transaction. The security of this digital money depends upon a single private key, which, if stolen, leads to a quantifiable dissipation of finances.

**Case study of Cryptocurrencies** A significant number of breaches of wallets are well-known. Mt Gox[1] being the most famous bitcoin (BTC) hack in which a humongous amount of BTC was stolen that completely sank Mt. Gox in 2014. It was known that the hacker had a hold of auditor credentials of Mt. Gox through the cryptowallet and transferred the digital money to an address for which the organization had no keys. Bitfinex was the second largest Bitcoin hack after Mt.Gox because the hackers got the advantage of vulnerability in its multi-signature bitcoin wallet architecture. The breach claimed Poloneix and Bitstamp are some other cryptocurrency hacks where either the private keys or the crypto wallets were hacked [2].

[1] https://blockonomi.com/mt-gox-hack/.

### 2.2 51% Attack

A 51% attack on a blockchain network refers to a miner or a sub-network of miners trying to obtain the control of more than 50% of hash rate or computing power of the network's mining [10]. The people in control of such mining power prevent new transactions from gaining confirmations on the network and get halted. Since transactions on a blockchain network is confirmed only after all the nodes on the network are in agreement. Until then the transaction is unverified. This time delay creates a perfect vector for the cyber attacks. Double spending is a common blockchain attack that exploits the verification mechanism of a transaction. The attackers trick the system to use the same cryptocurrencies in other transactions. Sometimes the goal of the attackers isn't always to double spend cryptocurrencies, but more often to cast discredit by affecting the integrity of a certain blockchain or a cryptocurrency.

### 2.3 User Wallet Attacks

Blockchain and cyber security go hand in hand; but sometimes blockchain users pose the greatest security threat. People overestimate the security protocol of blockchain and fail to observe its possible attack vectors. User wallet credentials are the principal target for cyber criminals. Hackers utilize the traditional methods like phishing (obtaining sensitive information such as usernames or passwords by disguising itself as a trustworthy entity) and dictionary attacks (breaking the cryptographic hash of the victim to find the wallet credentials) as well as the advanced techniques to locate the weak points in the cryptographic algorithms.

The digital signatures used in a blockchain are created by using various cryptographic algorithms. For instance, Bitcoin uses ECDSA cryptographic algorithm to generate public and private keys. But due to the insufficient entropy of the ECDSA, values can get repeated in more than one signature thereby providing loopholes to the attackers. The cold wallets (hardware wallets) and hot wallets (software applications) are also vulnerable to attacks by the adversaries. The hackers can obtain the PINs as well as the public/private keys and other sensitive information by breaking into the wallets..

## 3 Quantum Computing

Quantum computing harnesses the astonishing laws of quantum mechanics to process information and to perform computation. A quantum computer works on 'qubits' or quantum bits rather than the traditional 'bits' used in the classical computers. A bit encodes either a zero or one at any given time whereas a qubit can be zero or one or a superposition of the two qubit states simultaneously. This happens due to the quantum behavior of qubits; hence we can capitalize the superposition and entanglement phenomena of quantum physics. There is no doubt that quantum computers are going to be way too faster than our classical computers as they have the advantage of using ones, zeros and superposition of ones and zeros [11].

Quantum superposition is the fundamental principle of quantum mechanics. Schrodinger's theory of probability permits the existence of two or more waves simultaneously which leads to a potentially unlimited number of superposed waves. This implies that microscopic particles can theoretically be located in unlimited places at once, and can behave in possibly unlimited different ways. The principle of quantum superposition states that the most general state of a physical system, which can be in one of its many possible configurations, is the combination of all of these possible configurations given by the equation:

$$\alpha|0> + \beta|1>$$

where the coefficients $\alpha$ and $\beta$ are the complex numbers describing how much amount goes into each configuration if the physical system under consideration has two possible states 0 and 1 (0 for spin down and 1 for spin up of a qubit respectively). The probability for a specified configuration is given by:

$$P_0 = |\alpha|^2$$
$$P_1 = |\beta|^2$$
$$P_0 + P_1 = 1$$

Since, the probabilities should add up to one; the electron is in one of the two states for sure. Quantum computers have capitalized this ability of an atom/particle to be in a superposition of states to produce substantially increased power and much higher speed of computations.

## 4 Quantum Teleportation and Entanglement

Teleportation is an interesting phenomena of quantum mechanics in which quantum information possessed by any particle can be transmitted from one location to another using quantum entanglement between the communicating parties and a classical communication channel. China has been successful in teleporting a quantum state using a photon on the earth to a satellite which is approximately 1400 kms away. Teleportation utilizes quantum entanglement as a transmission medium which is used to teleport a quantum state from one location to another. When there are two or more particles are entangled, their quantum states become interdependent, irrespective of the distance between them. They behave as a single quantum object which can be described by a single wave function [12,13].

Teleportation is possible only because of quantum entanglement. Quantum entanglement is a strange spin-off of quantum mechanics which says that two particles can instantly communicate with each other, even across cosmic distances [14]. An entangled system can be

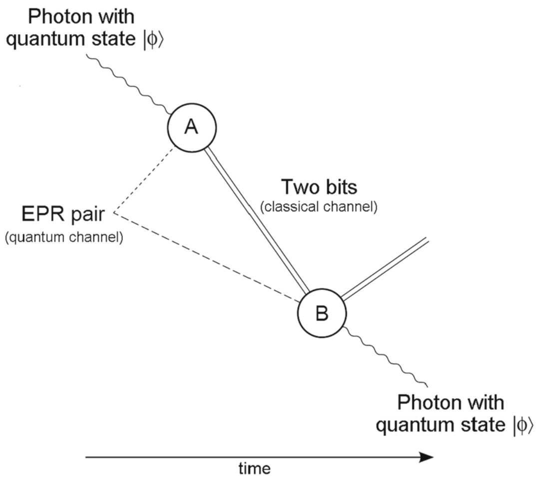
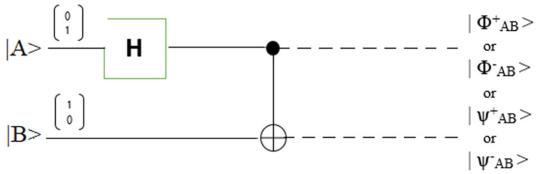



**Fig. 2** Quantum teleportation through EPR pair

**Fig. 3** Four orthogonal Bell states




described as the one where the quantum state on one particle cannot be determined without considering the quantum state of the other i.e the particles hold mutually exclusive states and hence are inseparable [15] (Fig. 2). This leads to the correlation between the observable physical properties of the system. The state of a composite system is the sum, or superposition, of products of its states of local constituents; but in an entangled system, the sum essentially consists of more than one term [15]. Two particles remain entangled until and unless there is some kind of interaction with the environment. For instance, if a measurement is made on any of the entangled particles, the entanglement is broken.

### 4.1 How Does Teleportation Work?

The process involves sharing an entangled pair of particles between the communicating parties. A joint measurement is made on the state of the particle that sender needs to teleport to receiver along with the sender's pair of entangled particle. The individual states of the entangled particles are never measured. Based on the outcome of that measurement, which is sent to sender via a classical communication channel, receiver performs a small number of simple operations to the other entangled particle, which leaves it in exactly the state of the original particle that was being teleported.

The key to the process is that the entangled pair of particles do not have well-defined states, but are correlated in a non-local fashion. It does not send a set of random numbers with definite values from one place to another; instead it is sending an indeterminate state whose exact value will not be determined until it is measured, at which point its value will absolutely determine the state of a second particle a long distance away (Fig. 3).

Recently IBM launched an integrated quantum computer system for scientific and commercial use called as IBM Q System One[2]. Since quantum era is destined to take over the classical computing system within a decade, we propose a more secure method for performing transactions in a blockchain using the teleportation phenomenon of quantum mechanics. We will exploit this entanglement to generate highly secure public keys to be used for a transaction in a blockchain. Hacking the teleported public key is impossible because of the spooky nature of qubits that makes them collapse whenever a measurement is performed on them by any third party. Before diving into the proposed solution directly, we need to mention some of the concepts here and the basic computations behind them.

## 5 Quantum Digital Signatures

The Blockchain technology makes extensive use of public key cryptography which works on the principle of a function that is not reversible in which computing the outcome is easier but the reverse is highly difficult. A quantum secure theme is proposed to implement on

[2] https://www.research.ibm.com/ibm-q/system-one/.

the blockchain technology using Quantum Digital Signatures which also uses asymmetric keys. The scheme allows the sender to create at least one pair of the private key and the equivalent public keys. The sign can then be validated by one or more recipients for the consensus of agreement or that the message has been forged. We can define a quantum one way function that delineates identical properties as the classical scheme but computing the reverse function is impossible even with the most powerful artifice techniques. This section introduces quantum one-way functions rooted in the principles of quantum mechanics [16].

### 5.1 One-Way Function

f(x) is a one way function if

- For any x, f(x) can be computed easily.
- For any f(x), x is difficult to compute.

Here we are using the fingerprint scheme of quantum one way function proposed by Gottesman [16].

### 5.2 Quantum One-Way Function

Earlier it has been introduced in [16] where the input is a classical bit-string $x$ of length L, where $L = O(2^n$ with $\partial \approx 0.9$ and the output is a quantum state $|f_x>$ of n qubits, then $f : |x>_a \rightarrow |f(x)>_b$ is a one way function if :

- There exists a polynomial time algorithm that computes $|f(x)>$ for any given $|x>$.
- There exists no possible algorithm that computes $|x>$ for any given $|f(x)>$.

where $a \gg b$. The states are nearly orthogonal: $|<f_x \mid f_{x'}>| \leq \partial$ for $x \neq x'$ which allows L to be very large than $n$. A Quantum one way function is provided with a quantum output state, with which one cannot extract enough information about the input string to reproduce it without knowing the $x$ due to a fundamental theorem of quantum information theory. According to Holevo's theorem, given an n-qubit quantum state, one cannot withdraw more than n classical bits of information as it establishes an upper bound on the amount of information that can be extracted from a given quantum state.

### 5.3 Holevo's Theorem

Given M copies of the state $|fx>$, at most $M_n$ bits of information about x can be retrieved when $L - M_n \gg 1$. We can use nearly orthogonal states to ensure the usage of less qubits for a string of bits that is of a certain length giving us the feasibility to generate a basis with more than two states. Therefore, to describe an information of $2^n$ bits, only $m$ qubits are reqiured to represent $n$ classical bits when $3^m = 2^n$ holds. Owing to Holevo's theorem and the fact, that $m$ should be much smaller than $n$, one can retrieve only $m$ bits out of the $n$ bit message. In the easier terms, we can say that if one gets $K$ copies of the public key, they can extract at most $Km$ bits of the private key [17]. From the above definitions and discussions, we can conclude that $|x> \rightarrow |f(x)>$ acts as a one way function where the probability of guessing the string x remains very small.

### 5.4 Distributing the Public Key

The quantum states work on the no cloning theorem, i.e. information of a quantum state cannot be copied. Unlike the existing public key infrastructure in which a copy of the public key is available to everyone on the network; this is not possible with quantum states. Therefore, the public keys can only be created and distributed by the person who knows the exact quantum state generally the sender since he is the one who knows the private key. Also, there is an upper bound on the number public quantum keys that can be created for a transaction. We are using Quantum Teleportation here for the distribution of public key to a limited number of recipients (miner nodes).

### 5.5 The Swap Test

Since the public keys are not copied they have to be sent to the miners on the blockchain individually via teleportation. Now we have to make sure that public keys distributed to the miner nodes are same so that every miner gets identical results while testing for the authenticity of the message. To do this, the miners need to perform the swap test.

## 6 Methodology

This section describes the procedures used for performing a blockchain transaction. The Bell measurement procedure that is required to generate entangled EPR pair of particles for teleportation and the swap test procedure for comparing the keys of the sender and the receiver are described here considering a general state for both. The illustration of Bell measurement is as follows:

$$|\Phi^{+}_{AB}>=\frac{1}{\sqrt{2}}(|0_A 0_B>+|1_A 1_B>)$$
$$|\Phi^{-}_{AB}>=\frac{1}{\sqrt{2}}(|0_A 0_B>-|1_A 1_B>)$$
$$|\Psi^{+}_{AB}>=\frac{1}{\sqrt{2}}(|0_A 1_B>-|1_A 0_B>)$$
$$|\Psi^{-}_{AB}>=\frac{1}{\sqrt{2}}(|0_A 1_B>-|1_A 0_B>)$$

Unknown state of the first particle can be written as:

$$|\Phi_A>=\alpha|0_A>+\beta|1_A>$$

where $|\alpha|^2+|\beta|^2=1$

The status of the three particles before sender performs measurement are

$$\begin{aligned}|\Psi^{+}_{ABC}> = &\frac{\alpha}{\sqrt{2}}(|0_A>|0_B>|1_B>-|0_A>|1_B>|0_C>)\\ &+\frac{\beta}{\sqrt{2}}(|1_A>|0_B>|1_C>-|1_A>|1_B>|0_C>)\end{aligned}$$

The above equation in Bell states can be written as

$$\begin{aligned} |\Psi^{+}_{ABC} > = \frac{1}{2}[|\Psi^{-}_{A} B > (\alpha|0_C > -\beta|1_C >) + |\Psi^{+}_{A} B > (\alpha|0_C > -\beta|1_C) \\ + |\Phi^{-}_{A} B > (\alpha|1_C > +\beta|0_C >) + |\Phi^{+}_{A} B > (\alpha|1_C > -\beta|0_C >)] \end{aligned} \tag{1}$$

### 6.1 Bell Measurement

The Bell measurement is performed with two EPR pairs of entangled particles. Sender and receiver are distributed the photon pairs generated using a parametric down conversion source. They both possess two sets of channel measurement devices to read and measure the channels. Now, sender and receiver performs Bell measurement on their EPR particle. The measurement is carried out on Bell operator orthonormal basis which considers four maximally entangled Bell states:

The outcome of the four states are equal likely with probability $\frac{1}{4}$. Particle C will take one of the four superimposed pure states given by equation 1. In this way, teleportation is achieved given that sender transmits the classical outcome to receiver after performing the measurement. Then, receiver makes the required transformation by performing the rotation operation.

### 6.2 Swap Test

The swap test is performed at the receivers end after receiving the key pairs from the sender. To perform the test, miner has to compare the following:

$$|f'_k >= |f_k >$$

– Swap Test Procedure: A quantum circuit is required for performing the swap test that comprises of a Fredkin Gate, A Hadamard gate and an ancilla qubit.
– Fredkin Gate: It is a controlled circuit with three inputs and three outputs. The output of the controlled swap gate includes the first bit, unchanged and last two bits are swapped, iff, the first bit is 1 [18].
– Hadamard Gate: It works on a single qubit to create a superposition by mapping the basis state $|0 >$ to

$$\frac{1}{\sqrt{2}}(|0 > +|1 >)$$

and $|1 >$ to

$$\frac{1}{\sqrt{2}}(|0 > -|1 >)$$

implying the equal probabilities of 0 and 1 for the measurement of the said qubit.
– Ancilla Qubit: Used to store the entangled states of the qubits.

The following quantum circuit proposed by Gottesman [16] is used to compare the two states.

Following the procedure described in Fig. 4, firstly, the Ancilla qubit $a$ is set to one of the symmetric state say

$$\frac{1}{\sqrt{2}}(|0 > +|1 >)$$

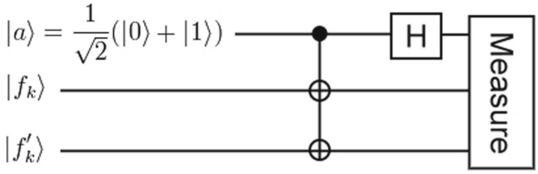



**Fig. 4** A pictorial representation of the swap test procedure

Secondly, this state is then used as a control on the target functions $|f_k^{'}>$ and $|f_k>$ using a Fredkin Gate. Lastly, a Hadamard gate is applied on $a$ to measure the first qubit. If the two states are same, $|0>$ is measured as the result and if the two states are almost orthogonal, the result can be $|0>$ or $|1>$. We outline the Swap test computation. The overall state of the qubit can be given by

$$|\Phi_0>=|a>|f_x>|f_x'>$$

Expanding the equation using $a$:

$$|\Phi_0>=\frac{1}{\sqrt{2}}(|0>+|1>)|f_x>|f_x'>$$

$$|\Phi_0>=\frac{1}{\sqrt{2}}(|0>|f_x>|f_x'>+|1>)|f_x>|f_x'>)$$

Apply the Fredkin gate in above equation:

$$|\Phi_0>=\frac{1}{\sqrt{2}}(|0>|f_x>|f_x'>+|1>)|F_x>|F_x'>)$$

Similarly, we apply Hadamard Gate to first qubit:

$$\begin{aligned}|\Phi_0>=&\frac{1}{2}[(|0>+|1>)|f_x>|f_x'>\\&+(|0>-|1>)|f_x>|f_x'>]\end{aligned}$$

On sorting $|0>$ or $|1>$, the overall state becomes:

$$\begin{aligned}|\Phi>=&\frac{1}{2}[|0>(|f_x>|f_x'>+|f_x'>|f_x>)\\&+|1>(|f_x>|f_x'>-|f_x'>|f_x>)]\end{aligned} \tag{2}$$

Now from the Eq. (2), it is easy to compare that if the states $|f_x>=|f_x'>$, then $|\Phi>=|0>|f_x>|f_x>$, which gives a equals to 0 when it is measured. And a measurement result of 0 implies that the qubit reached the receiver intact.

## 7 The Proposed Algorithm

Consider a simple communication scenario on a blockchain where A wants to send a message to B. The sender A has to sign each and every bit of the message (i.e., $m \in \{0, 1\}$) to be sent because in case of quantum digital signatures, hash will not work. The Algorithm consists of three sections namely *signing* that includes the key generation and signing of the message with the quantum digital signature. The second section consists of the distribution of the keys to the network using the teleportation. The third and final segment consists of the validation process that verifies the sender's and the receiver's key (Fig. 5).

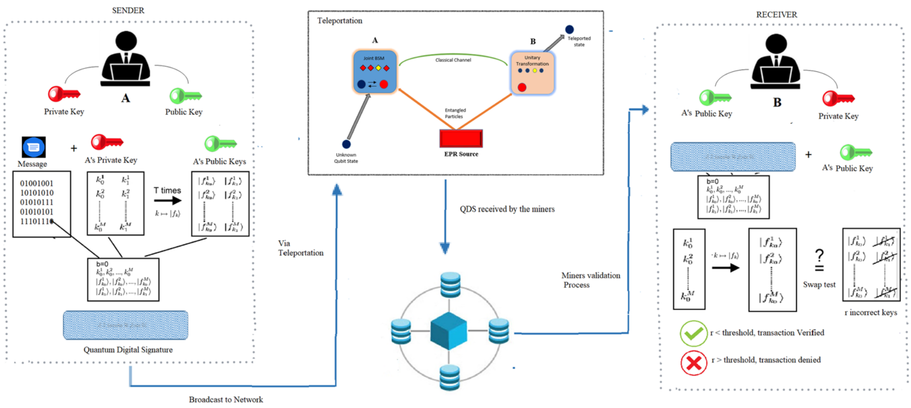



**Fig. 5** The Proposed Scheme; (1) *Signing* of the keys and generation of quantum digital signature at the sender's end. (2) *Distribution of the keys via teleportation* using EPR pairs for secured transactions in blockchain. (3) *validation* at the receivers end, in this case, receivers are the miner nodes on the blockchain network

## 7.1 Signing

The following steps comprise the key generation for the transaction:

- Sender chooses P pairs of private keys $\{k_0^i, k_1^i\}$, where $1 <= i <= P$
- All the $k_0$ keys will be used to sign the message-bit $m$ if $m = 0$ and all the $k_1$ keys will be used to sign the message-bit if $m = 1$.
- The function that maps $k \rightarrow f_k$ is known to all the nodes on the network. Sender computes the corresponding public keys $\{f(k_0^i), f(k_1^i)\}$ and teleports all of them to the miner nodes.

The level of security permits a limited number for creation of identical public keys.

## 7.2 Distribution of the Keys via Teleportation

The essentials for quantum teleportation on a blockchain are a qubit to be teleported, a classical communication channel with the capability of transmitting two classical bits, a medium to generate and transport EPR pair of entangled qubits to two different nodes in a blockchain and creation of the bell states by performing bell measurement on one of the EPR pair. The quantum teleportation process for Blockchain requires following steps:
**Initialization**:

- Identify the Sender node, say $N_1$ and the receiver node, say $N_2$.
- $N_1$ generates quantum digital signature for the transaction.
- Key pair qubits are sent to the Quantum circuit at node $N_1$.

**Transaction Execution at the receiver's end:**

- Retrieve the Key pairs at Node $N_2$ (receiver) using the qubit information received from $N_1$.
- Validate using the Swap test.

**do**

1. Perform Bell measurement on the EPR pair qubit and the key qubit that is to be teleported.
2. One of the four bell measurement outcomes is generated.
3. Encode the generated Bell state into two classical bits of information.
4. Discard the qubits at node $N_1$
5. Send the classical bits generated in (3) using a classical communication channel from node $N_1$ to node $N_2$.
6. At node $N_2$ (Receiver):
   (a) The EPR pair qubit at node $N_2$ is in one of the four bell states due to the measurement performed at node $N_1$.
   (b) Decode the information received in the two classical bits from node $N_1$ and identify the original quantum state.
7. The resultant qubit is the original quantum state of the qubit that was to be teleported.

**while** key_qubit ! = null

## 7.3 Validation by the Receiver

The receiver (miner node) currently has the message-bit m, the relative private keys $k_0$ or $k_1$ and all public keys $\{f(k_0^i), f(k_1^i)\}$

1. Miner calculates $f|k_b>$ for all received private keys $k_0$ or $k_1$.
2. Execute the swap test.
3. Measure the calculated states of the key with the received public keys. The miner node needs to do the swap test for all the keys and count the number of incorrect keys. Let it be $t$.
   (a) If the count of incorrect keys is small, then the bit is most probably valid.
   (b) If the count of incorrect keys is large, then discard the message. When $t$ is large, Miner can conclude with high probability that the message has been forged by somebody.
4. In case of (a), continue with the message verification process and add the transaction to the blockchain. If the case is (b), then discard the message and inform the sender.
5. Go to *Signing*.

The threshold $t$ for incorrect keys is a matter of concern here as there are several recipients and all of them will measure $t$ with different probabilities. So a single threshold value cannot serve the purpose here leading to invalidation of a message by some of the recipients. To prevent this, more than one threshold values need to be defined. This can be prevented by defining more than one threshold as follows:

– Accept if $T_a = aP$, the number of incorrect keys is below $T_a$, the bit is highly valid.
– Reject if $T_b = bP$, the number of incorrect keys is above $T_b$, the bit is forged.

If the value of the threshold lies in between the accept and reject thresholds, then the recipient cannot be sure that he validated the message properly or not.

## 7.4 Resistance of the Algorithm Against Attacks

Since the proposed algorithm facilitates the transaction processing in a Blockchain, the resistance against the major attacks is one big issue that needs to be discussed in lieu of the proposed algorithm. The double spend attack requires the information regarding the transaction that has been executed once. The attack can be successful if any third party is able to acquire the information of the transaction processed earlier. Based on that information, the attacker would try to gain benefit from the same transaction again without actually spending the money. But , the proposed algorithm uses Quantum Digital Signature scheme for the execution of transactions; therefore the attacker needs the digital signature to perform the transaction. The signature once generated for a particular transaction is valid only for that transaction and as soon as the receiver receives all the qubits and decodes the information sent via the classical channel, the information shared amongst the sender and receiver is destroyed because of the quantum entanglement between the particles and there is no other way to get that information again. Hence, the adversaries trying to attack the blockchain and wish to execute similar transactions again would never be able to do so.

### 7.5 Addressing the Scalability Issue of Blockchain

The scalability issue is related to the number of transactions that can be executed on a blockchain network . This has been the biggest issue since the evolution of the blockchain technology. The frequency of creating a block on a blockchain is estimated roughly about 10 minutes. Also the blocksize is limited to 1 megabyte. These are the two major limitations that constrain the throughput of the blockchain network. A number of solutions have been proposed to tackle these issues including "fork", Schnorr signatures and Merkelized abstract Syntax trees to name a few. Several proposals for increasing the block size limit have also been made. But the scalability issue still pertains because of the distributed nature of the blockchain.

The Algorithm that has been proposed in this paper would be able to address this issue effectively. The reason being the usage of the Quantum Teleportation for transferring the information from sender to the receiver. The information sent through teleportation is received instantaneously by the receiver because of the entanglement between the particles of the sender and receiver. The word "instantaneous" here means that the information is travelling at the speed of light , but since teleportation requires a classical channel, so we cannot say that the information will travel at the speed of the light. But it will be much faster than the fiber optic cable which slows down the speed of the light waves travelling through it by approximately 30%. If we consider an average transaction size , the maximum possible transactions per unit time on a blockchain are 4–7. If we use the proposed algorithm, the average number of transactions would be increased. Since the quantum digital signature would be teleported instantaneously to all the nodes working on the blockchain; only delay would be for the matching of the key pairs. The proposed algorithm , if used in collaboration with an increased block size would surely decrease the transaction processing time thereby addressing the scalability of the network.

## 8 Discussion

The state of the sender's original qubit is never directly measured in quantum teleportation. This means that the qubit arrives completely intact at the receiver's end. This is the only sure way of transmitting an unknown quantum state without sending the particle in that state directly. Also, the "no-cloning" theorem forbids creating a copy of an unknown quantum state. Since teleportation uses the classical communication channel, there may be some eavesdropper trying to access the information sent via this medium. But due to the fundamental aspect of quantum mechanics, any third party or adversaries trying to gain knowledge about the state of the qubit must somehow measure it thereby introducing anomalies in the system and ultimately breaking up the entanglement.

This paper presented the transmission of QDS on a blockchain network where sender and receiver are on the same network using the EPR pair amongst the communicating parties. But in real time, there can be multiple sender/receiver transactions at the same time on a blockchain network. With the dawn of quantum internet, it becomes essential to establish a quantum as well as digital communication channel on a blockchain so that any number of nodes can transact simultaneously. This initiates the need of quantum routers (required for digital communication of Bell state measurement results) and quantum repeaters (required for entanglement swapping amongst multiple EPR pair on a larger network) for successful and secure transmission of the teleported state.

## 9 Conclusion and Future Work

Blockchain is a relatively a new technology with its pros and cons. It also has bugs and vulnerabilities that are exposing it to an increased number of cyber intrusions. The existing public key infrastructure is susceptible to attacks. A model of highly protected blockchain transaction mechanism utilizing the quantum teleportation phenomenon has been proposed. We have presented a transaction mechanism based on quantum digital signatures that are transported via teleportation for enhanced security of the key pairs. The QDS scheme utilizes the swap test for validation of the information received and teleportation denies the possibilities of copying of the quantum information transmitted via quantum channels. The two level verification in our algorithm doubles the security of the transaction in a blockchain hence providing highly secure transactions. The proposed algorithm for teleporting quantum digital signatures (QDSs) on a blockchain has generated opportunities for creating a more secure and faster Blockchain. This work will contribute in the creation of safer public and private blockchains and will initiate further applications of quantum digital signatures in securing the public key infrastructures.

Recently Wehner et al. [19] investigated about *Quantum Internet* which when applied to the blockchain technology will create a whole new quantum-like blockchain capable of executing the most efficient, secure and rapid transactions. Also, teleportation is the most secure possible way of transmission as it abides by the "no-cloning" theorem. Hence the security of a blockchain shall be enhanced exceptionally and will prevent any future attacks on this technology. Quantum information is also susceptible to noise and decoherence that can generate unwanted errors in the message. For the futuristic blockchain technology, quantum error correction (QEC) codes can be used to protect the information travelling on quantum channels from such errors. QEC's are essential for achieving fault-tolerant quantum computation that safeguards the quantum information from noise, identifies erroneous quantum gates, invalid quantum state preparation and inaccurate measurements.

Our future work will be to simulate the proposed algorithm for multiple sender/receiver transactions using the IBM Q simulator[3] available for research purposes on the cloud. The authors look forward to creating a quantum blockchain that shields the privacy, boost up the security and minimize the latency while performing any transaction with any number of nodes. This research holds promise for providing secure blockchain in the future quantum era.

[3] https://www.research.ibm.com/ibm-q/.

## References

1. Nakamoto S (2008) Bitcoin: a peer-to-peer electronic cash system (2008)
2. Tschorsch F, Scheuermann B (2016) Bitcoin and beyond: a technical survey on decentralized digital currencies. IEEE Commun Surv Tutor 18(3):2084–2123
3. Chaum D (1983) Blind signatures for untraceable payments. Advances in cryptology. Springer, Boston
4. Wan J et al (2019) A blockchain-based solution for enhancing security and privacy in smart factory. IEEE Trans Ind Inf 15:3652–3660
5. Qian J, Tiwari P, Gochhayat SP, Pandey HM (2020) A noble double dictionary based ECG compression technique for IoTH. IEEE Intern Things J. https://doi.org/10.1109/jiot.2020.2974678
6. Kumar S, Tiwari P, Zymbler M (2019) Internet of Things is a revolutionary approach for future technology enhancement: a review. J Big Data 6(1):111
7. Bebortta S, Senapati D, Rajput NK, Singh AK, Rathi VK, Pandey HM, Jaiswal AK, Qian J, Tiwari P (2020) Evidence of power-law behavior in cognitive IoT applications. Neural Comput Appl. https://doi.org/10.1007/s00521-020-04705-0
8. Gochhayat SP, Kaliyar P, Conti M, Tiwari P, Prasath VBS, Gupta D, Khanna A (2019) LISA: lightweight context-aware IoT service architecture. J Clean Prod 212:1345–1356
9. Peters G, Panayi E, Chapelle A (2015) Trends in cryptocurrencies and blockchain technologies: a monetary theory and regulation perspective. J Financ Perspect 3(3)
10. Courtois NT, Bahack L (2014) On subversive miner strategies and block withholding attack in bitcoin digital currency. arXiv preprint arXiv:1402.1718
11. Chen G et al (2006) Quantum computing devices: principles, designs, and analysis. Chapman and Hall/CRC, Boca Raton
12. Riebe M et al (2004) Deterministic quantum teleportation with atoms. Nature 429(6993):734
13. Barrett MD et al (2004) Deterministic quantum teleportation of atomic qubits. Nature 429(6993):737
14. Bennett CH et al (1993) Teleporting an unknown quantum state via dual classical and Einstein–Podolsky–Rosen channels. Phys Rev Lett 70(13):1895
15. Deutsch D, Hayden P (2000) Information flow in entangled quantum systems. Proc R Soc Lond Ser A: Math Phys Eng Sci 456(1999):1759–1774
16. Gottesman D, Chuang I (2001) Quantum digital signatures. arXiv preprint arXiv:quant-ph/0105032v2
17. Nielsen MA, Chuang I (2002) Quantum computation and quantum information. Am J Phys 70:558–559
18. Brown J (2001) Quest for the quantum computer. Simon and Schuster, New York
19. Wehner S, Elkouss D, Hanson R (2018) Quantum internet: a vision for the road ahead. Science. https://doi.org/10.1126/science.aam9288



## Affiliations

**Sheetal Singh[1] · Nikhil Kumar Rajput[1] · Vipin Kumar Rathi[1] · Hari Mohan Pandey[2] · Amit Kumar Jaiswal[3] · Prayag Tiwari[4]**

✉ Nikhil Kumar Rajput
nikhil45_scs@jnu.ac.in

✉ Vipin Kumar Rathi
vipin68_scs@jnu.ac.in

✉ Prayag Tiwari
prayag.tiwari@dei.unipd.it

Sheetal Singh
sheetal.singh@ramanujan.du.ac.in

Hari Mohan Pandey
pandeyh@edgehill.ac.uk

Amit Kumar Jaiswal
amitkumar.jaiswal@beds.ac.uk

[1] Department of Computer Science, Ramanujan College, University of Delhi, New Delhi 110019, India

[2] Department of Computer Science, Edge Hill University, Ormskirk, UK

[3] Institute for Research in Applicable Computing, University of Bedfordshire, Lancashire, UK

[4] Department of Information Engineering, University of Padova, Padova, Italy